

DSA Day 13: Queue & Circular Queue

Book: Data Structures Through C in Depth - Ch4 §4.2-4.3 · pp 114-123 | Code in Python

Overview

A queue is FIFO. A plain array queue wastes space as front advances; a circular queue reuses slots with modulo arithmetic.

Key concepts to master

- front/rear
- wrap-around with %
- full vs empty detection

Python implementation

```
class CircularQueue:
    def __init__(self,k):
        self.q=[None]*k; self.k=k; self.front=0; self.size=0
    def enqueue(self,x):
        if self.size==self.k: return False
        self.q[(self.front+self.size)%self.k]=x; self.size+=1; return True
    def dequeue(self):
        if not self.size: return None
        v=self.q[self.front]; self.front=(self.front+1)%self.k; self.size-=1; return v
```

Complexity

enqueue/dequeue $O(1)$.

Common pitfalls

- Confusing full vs empty when `front==rear`
- Forgetting modulo

Practice

- Wrap a size-5 circular queue
- Why does a plain array queue waste space?

